



PROJECT-BASED LEARNING

PHASE-I EVALUATION

**AI-Powered Semantic Search for Intelligent
File Retrieval**

Team ID: 312

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID: 312
Semester: 5th
Section: E
Domain: *OS+DBMS*

Team Members

1. *Harsh Chahal – 240211874*

Mentor

(prof.)Dr. Sumit Pundir

Interactions completed: _____

Problem Statement & Motivation

Problem Statement

Traditional file systems rely on filenames, paths and exact keywords, making it difficult to retrieve a file when the user remembers its content but not its exact wording or location.

Motivation

- ▶ *Users often remember the content but forget the filename*
- ▶ *Keyword search cannot understand synonyms and context.*

Target Users / Stakeholders

- ▶ *Students and learners studying OS and DBMS.*
- ▶ *Developers and researchers.*

Objectives

Project Objectives

1. *Develop a virtual file system with directories and disk blocks.*
2. *Implement FAT/i-node based file allocation.*
3. *Integrate OS,DBMS, and NLP Concepts into a single retrieval Pipeline.*
4. *Evaluate Search Accuracy against traditional keyword based search.*
5. *Store Metadata and Embeddings in a relational database with scope for future Scaling.*

Key Objective: *To retrieve files based on semantic meaning rather than exact filenames or keywords.*

Proposed Solution

Proposed Approach

1. *Upload file.*
2. *NLP Algorithms.*
3. *Disk Scheduling Algorithm.*
4. *MySQL*
5. *Search Results*

Key Features

- ▶ *Semantic Search.*
- ▶ *Sentence Embedding.*
- ▶ *Virtual File System.*
- ▶ *MySQL.*
- ▶ *Metadata+vectors*

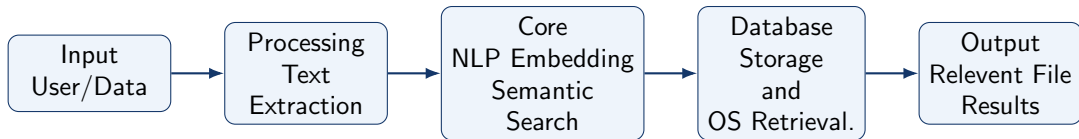
An intelligent search layer connects NLP-based semantic matching with database metadata and OS-level file retrieval.

Integration of PBL Course Concepts

Course	Concepts Applied	Project Module
Operating Systems	<i>Disk Storage,Block Allocation,Disk Scheduling,File permissions.</i>	<i>Virtual File System Module</i>
DBMS	<i>Metadata Storage, vector Embedding, indexing, Query retrieval.</i>	<i>Database Module</i>
AI/NLP	<i>Sentence-transformer Model(all-MiniLM-l6-v2.</i>	<i>NLP/Semantic Search Module</i>

Requirement: OS:Disk scheduling and file allocation are used for virtual file storage.
DBMS:Metadata and Vector Embedding are stored and indexed in a relational database.
AI/NLP:Sentence Embedding and Cosine Similarity are used for Semantic matching.

System Architecture / Workflow



Major Components

- ▶ *OS Module*
- ▶ *NLP Module*
- ▶ *DBMS Module*

Data / Control Flow

- ▶ *File upload→Text extraction→Embedding→database.*
- ▶ *Query→Embedding→Similarity search→ACL Check.*
- ▶ *Relevant files returned to user.*

Technology Stack & Methodology

Technology Stack

- ▶ Programming: *C / C++ / Python/ Other*
- ▶ Database: *SQLite / PostgreSQL*
- ▶ Tools / Framework: *Sentence Transformers/C++/SQLite3*
- ▶ IDE: *Visual Studio Code.*
- ▶ Version Control: *Git / GitHub*

Development Methodology

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Evaluation

C++ was chosen for OS level simulation; Python was chosen for NLP duw to library support; SQLite offers a lightweight database for local testing.

Initial Progress & Team Contribution

Progress Achieved

- ▶ *Project idea and requirements identified.*
- ▶ *OS, DBMS and NLP Concepts identified.*
- ▶ *System architecture and workflow designed.*
- ▶ *NLP and semantic search approach selected.*
- ▶ *SQLite selected for Database experiments and result Storage.*

Individual Contribution

Member	Role	Contribution
M1	Lead	Project coordination, system design and OS, DBMS, NLP module

Roadmap, Expected Outcomes & References

Roadmap

1. Phase-I: Proposal & Design
2. Phase-II: Development
3. Phase-III: Final Implementation

Expected Outcomes

- ▶ *Working Semantic Search Engine*
- ▶ *SQLite Performance Results.*
- ▶ *Virtual File System.*
- ▶ *Research in NLP.*

References

1. *Operating System Concepts* — Silberschatz et al.
2. *Fundamentals of Database Systems* — Elmasri & Navathe.
3. *Sentence Transformers* [sbert.net](https://www.sbert.net/).

Thank You
Questions & Suggestions